

# **ECON1050 Tutorial 1**

## **Sets, Logic & Algebra Review**

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# Today's Plan

- ▶ Intro
- ▶ Recap: core concepts from Lecture 1
- ▶ Tutorial questions: Q1–Q5

# A bit about Xavier Xi Zhang

- ▶ My name: Xavier Xi Zhang
  - ▶ You can call me **Xavier**, **Xav**, **Xi**, or **X**.
- ▶ Economics Honours student (ECON/MATH student before that).
- ▶ Personal website: [xavierxizhang.github.io](https://xavierxizhang.github.io)
  - ▶ Tutorial slides will be uploaded under **Teaching**.
- ▶ Questions are welcome (ECON, MATH, or beyond).
- ▶ My consultation hours are every Monday from 11 am – 12:30 pm, at 39-125B.

- ▶ This is one of the TWO maths courses offered specifically for ECON students.
- ▶ Three parts:
  - ▶ Foundations: sets, logic, algebra, solving equations, functions...
  - ▶ Calculus
  - ▶ Matrix algebra
- ▶ Some tips:
  - ▶ As long as you know how to solve all the tutorial questions, you should be able to get a 7.
  - ▶ So make sure you take the tutorials seriously!
- ▶ If you enjoy this course, I'd recommend taking more maths courses.

# Recap: Sets and notation

## Sets and elements

A **set** is a collection of elements. Membership is written as

$$a \in A \quad \text{or} \quad a \notin A.$$

A set may be listed using braces or described by a defining property.

## Subsets

$$A \subset B$$

means that every element of  $A$  belongs to  $B$ , and  $B$  contains at least one additional element. The empty set is denoted by  $\emptyset$ .

## Set-builder notation

$$\{x : x \text{ satisfies a stated property}\}.$$

# Recap: Set operations

## Intersection, union and difference

$$A \cap B = \{x : x \in A \text{ and } x \in B\},$$

$$A \cup B = \{x : x \in A \text{ or } x \in B\},$$

$$A \setminus B = \{x : x \in A \text{ and } x \notin B\}.$$

## Universal set and complement

If  $\Omega$  is the universal set, the complement of  $A$  is

$$A' = \Omega \setminus A.$$

Two sets are **disjoint** when  $A \cap B = \emptyset$ .

## Convex set

A set is convex when the line segment joining any two points in the set lies entirely inside the set.

# Recap: Propositions and logical relations

## Proposition

A proposition is a statement that is either true or false.

## Implication and converse

$$P \Rightarrow Q$$

means “if  $P$ , then  $Q$ .” Its converse is

$$Q \Rightarrow P.$$

The truth of an implication does not automatically imply the truth of its converse.

## Logical equivalence

$$P \Leftrightarrow Q$$

means that both  $P \Rightarrow Q$  and  $Q \Rightarrow P$  are true.

# Recap: Necessary and sufficient conditions

## When $P \Rightarrow Q$

- ▶  $P$  is a **sufficient condition** for  $Q$ .
- ▶  $Q$  is a **necessary condition** for  $P$ .
- ▶ Equivalent wording includes “ $P$  only if  $Q$ ” and “ $Q$  if  $P$ .”

## When $P \Leftrightarrow Q$

$P$  is a necessary and sufficient condition for  $Q$ , and vice versa. This is read as “ $P$  if and only if  $Q$ ” or “ $P$  iff  $Q$ .”



# Recap: Mathematical proofs

## Theorems and lemmas

Mathematical results are commonly stated as implications

$$P \Rightarrow Q,$$

where  $P$  contains the premises and  $Q$  is the conclusion.

## Proof methods

- ▶ **Direct proof:** derive the conclusion through a sequence of valid logical steps.
- ▶ **Proof by contradiction:** assume the required conclusion is false and derive a contradiction.
- ▶ **Mathematical induction:** verify a base case, assume the result for  $n = k$ , and prove it for  $n = k + 1$ .

# Recap: Algebraic rules

## Basic laws

$$\begin{aligned}a + b &= b + a, & ab &= ba, \\(a + b) + c &= a + (b + c), & a(bc) &= (ab)c, \\a(b + c) &= ab + ac.\end{aligned}$$

## Powers

For appropriate real numbers and integer exponents,

$$a^r a^s = a^{r+s}, \quad \frac{a^r}{a^s} = a^{r-s}, \quad (ab)^r = a^r b^r, \quad a^{-r} = \frac{1}{a^r}.$$

## Quadratic identities

$$(a + b)^2 = a^2 + 2ab + b^2, \quad (a - b)^2 = a^2 - 2ab + b^2, \quad (a - b)(a + b) = a^2 - b^2.$$

# Recap: Fractions and fractional powers

## Fractions

Denominators must be non-zero. The standard rules are

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}, \quad \frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}, \quad \frac{a/b}{c/d} = \frac{ad}{bc}.$$

## Fractional powers

$$a^{p/q} = \left(a^{1/q}\right)^p.$$

The symbol  $\sqrt{a} = a^{1/2}$  denotes the **principal square root**, which is non-negative.

# Recap: Inequalities

## Strict and weak inequalities

$$a > b, \quad a < b, \quad a \geq b, \quad a \leq b.$$

The statement  $a > b$  is equivalent to  $a - b > 0$ .

## Manipulation rules

- ▶ Adding or subtracting the same number preserves the direction.
- ▶ Multiplying or dividing by a positive number preserves the direction.
- ▶ Multiplying or dividing by a negative number reverses the direction.

# Recap: Sign diagrams and intervals

## Sign diagrams

To solve a product or rational inequality:

1. move all terms to one side and factor where possible;
2. identify zeros and points where the expression is undefined;
3. determine the sign on each interval;
4. include or exclude boundary points according to the inequality.

## Intervals

$(a, b)$ ,  $[a, b]$ ,  $(a, b]$ ,  $[a, b)$ .

Round brackets exclude an endpoint; square brackets include it.

# Recap: Absolute values

## Definition

For any real number  $a$ ,

$$|a| = \begin{cases} a, & a \geq 0, \\ -a, & a < 0. \end{cases}$$

The absolute value is the distance from the origin on the real number line.

## Absolute-value inequalities

For  $b > 0$ ,

$$|x| < b \Leftrightarrow -b < x < b,$$

$$|x| > b \Leftrightarrow x < -b \text{ or } x > b.$$

## Question 1

Use a sign diagram to determine for which values of  $x$  each inequality holds.

(a)

$$(x - 1)^3(x + 2) \leq 0.$$

(b)

$$2 < \frac{3x + 1}{2x + 4}.$$

# Tutorial Q1(a): Sign diagram

## Knowledge used

- ▶ A factor raised to an odd power has the same sign as the factor.
- ▶ Use zeros to divide the real line into sign intervals.
- ▶ Include zeros when the inequality is weak.

## Solution

The critical values are  $x = -2$  and  $x = 1$ . Since  $(x - 1)^3$  has the same sign as  $x - 1$ ,

| $x$                | $(-\infty, -2)$ | $-2$ | $(-2, 1)$ | $1$ | $(1, \infty)$ |
|--------------------|-----------------|------|-----------|-----|---------------|
| $(x - 1)^3$        | $-$             | $-$  | $-$       | $0$ | $+$           |
| $x + 2$            | $-$             | $0$  | $+$       | $+$ | $+$           |
| $(x - 1)^3(x + 2)$ | $+$             | $0$  | $-$       | $0$ | $+$           |

Therefore,

$$x \in [-2, 1].$$



# Tutorial Q1(b): Knowledge used

## Knowledge used

- ▶ Rearrange the inequality into one rational expression.
- ▶ Use numerator zeros and denominator zeros to divide the real line into sign intervals.
- ▶ A denominator zero is excluded from the solution.
- ▶ Do not multiply by an expression of unknown sign without checking whether the inequality reverses.

# Tutorial Q1(b): Solution

## Solution

$$2 < \frac{3x+1}{2x+4} \Leftrightarrow \frac{3x+1}{2x+4} - 2 > 0 \Leftrightarrow \frac{-x-7}{2x+4} > 0.$$

With critical values  $-7$  and  $-2$ ,

| $x$                 | $(-\infty, -7)$ | $-7$ | $(-7, -2)$ | $-2$      | $(-2, \infty)$ |
|---------------------|-----------------|------|------------|-----------|----------------|
| $-x-7$              | +               | 0    | -          | -         | -              |
| $2x+4$              | -               | -    | -          | 0         | +              |
| $\frac{-x-7}{2x+4}$ | -               | 0    | +          | undefined | -              |

Thus,

$$-7 < x < -2 \quad \text{or} \quad x \in (-7, -2).$$

# Tutorial Q2

## Question 2

Show that if

$$a + b > 0,$$

then

$$a > b \implies a^2 > b^2.$$

# Tutorial Q2: Direct proof

## Knowledge used

- ▶ Direct proof: begin with the assumptions and derive the conclusion.
- ▶  $a > b$  is equivalent to  $a - b > 0$ .
- ▶ The product of two positive numbers is positive.
- ▶ Difference of squares:  $(a + b)(a - b) = a^2 - b^2$ .

## Solution

From  $a > b$ ,

$$a - b > 0.$$

Together with  $a + b > 0$ , this gives

$$(a + b)(a - b) > 0 \Leftrightarrow a^2 - b^2 > 0 \Leftrightarrow a^2 > b^2.$$

## Question 3

For each implication, decide:

1. whether the implication is true;
2. whether the converse implication is true.

(a)

$$x = 5 \text{ and } y = -3 \implies x + y = 2.$$

(b)

$$(x - 3)^2(y + 2) > 0 \implies y > -2.$$

# Tutorial Q3(a): Implication and converse

## Knowledge used

- ▶ To verify an implication, apply its premises and check the conclusion.
- ▶ The converse of  $P \Rightarrow Q$  is  $Q \Rightarrow P$ .
- ▶ One counterexample is sufficient to show that an implication is false.

## Solution

The implication is true because

$$x + y = 5 + (-3) = 2.$$

Its converse is

$$x + y = 2 \Rightarrow x = 5 \text{ and } y = -3.$$

The converse is false:  $x = 7$  and  $y = -5$  give  $x + y = 2$  but do not satisfy the original premises. Hence,

the implication is true, but its converse is false.

# Tutorial Q3(b): Implication and converse

## Knowledge used

- ▶ A square is always non-negative; strict positivity also requires the squared factor to be non-zero.
- ▶ Test the converse separately and use a counterexample when it is false.

## Solution

Because  $(x - 3)^2 \geq 0$  for every real  $x$ ,

$$(x - 3)^2(y + 2) > 0 \Leftrightarrow x \neq 3 \text{ and } y > -2.$$

Hence the original implication is true. Its converse is false because  $x = 3$  makes the product equal to zero even when  $y > -2$ .

Original implication: true. Converse implication: false.

## Question 4

Fill each blank with “iff”, “if”, or “only if”, and use the corresponding implication sign.

(a)

$$x = \sqrt{4} \quad \boxed{\phantom{000}} \quad x = 2.$$

(b)

$$x(x+3) < 0 \quad \boxed{\phantom{000}} \quad x > -3.$$

(c)

$$x^2 < 9 \quad \boxed{\phantom{000}} \quad x < 3.$$



# Tutorial Q4: Knowledge used

## Knowledge used

- ▶ “ $P$  only if  $Q$ ” means  $P \Rightarrow Q$ .
- ▶ “ $P$  if  $Q$ ” means  $Q \Rightarrow P$ .
- ▶ “ $P$  iff  $Q$ ” means  $P \Leftrightarrow Q$ .
- ▶  $\sqrt{a}$  denotes the non-negative principal square root.
- ▶ Sign diagrams and interval notation identify the exact solution set of an inequality.

# Tutorial Q4: Answers

## Solution

(a) By the principal-square-root convention,

$$x = \sqrt{4} \Leftrightarrow x = 2.$$

Therefore the correct word is *iff*.

(b) Since

$$x(x+3) < 0 \Leftrightarrow -3 < x < 0,$$

we have  $x(x+3) < 0 \Rightarrow x > -3$ , but not conversely. Therefore,

$$x(x+3) < 0 \text{ only if } x > -3.$$

(c) Since

$$x^2 < 9 \Leftrightarrow -3 < x < 3,$$

we have  $x^2 < 9 \Rightarrow x < 3$ , but not conversely. Therefore,

$$x^2 < 9 \text{ only if } x < 3.$$

## Question 5

Let  $\Omega$  be the set of all students at a university. Let

$F$  = female students,  $M$  = mathematics students,  $C$  = choir members,

$B$  = biology students,  $T$  = tennis players.

Write the following statements in set terminology.

- (a) All biology students who are mathematics students.
- (b) There are female biology students in the choir.
- (c) No tennis player studies biology.
- (d) Students who neither play tennis nor belong to the choir all study biology.

# Tutorial Q5: Knowledge used

## Knowledge used

- ▶ “and” corresponds to intersection, while “or” corresponds to union.
- ▶ “neither  $A$  nor  $B$ ” corresponds to the complement of  $A \cup B$ .
- ▶ “there are” means the relevant set is non-empty.
- ▶ “no members” means the relevant intersection is empty.
- ▶ “all members of one set belong to another” is expressed using a subset relation.

## Solution

- (a)  $B \cap M$ .
- (b)  $F \cap B \cap C \neq \emptyset$ .
- (c)  $T \cap B = \emptyset$ .
- (d)  $\Omega \setminus (T \cup C) \subset B$ .